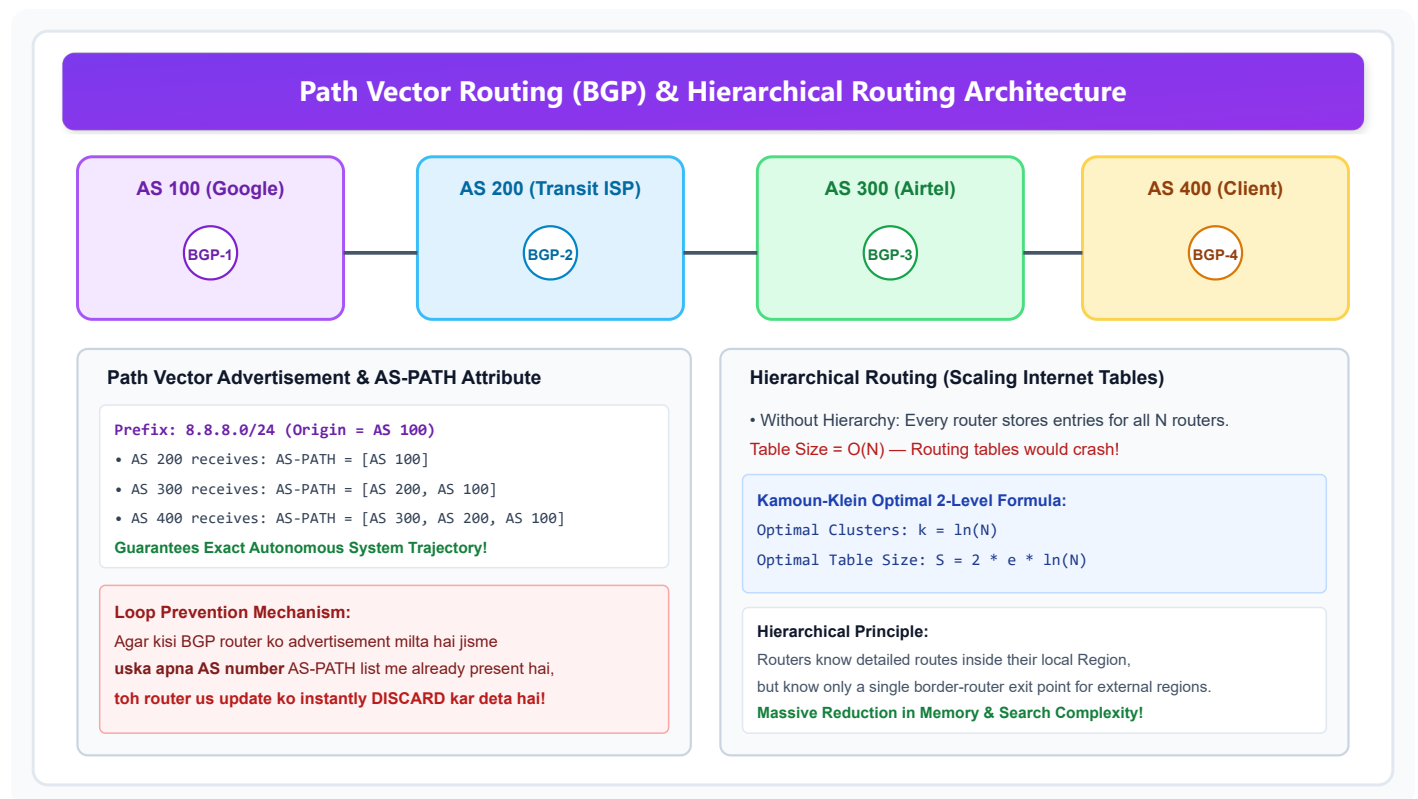


Module 13: Path Vector, Hierarchical Routing, and BGP

Unit 3: Network Layer

Inter-Domain & Hierarchical Scaling

Inter-Domain Routing Foundations, Border Gateway Protocol (BGP-4), Path Vector AS-PATH Attribute, Loop Prevention, and Kamoun-Klein Hierarchical Clustering.



1. BGP Path Vector & Loop Prevention

Path Vector Advertisement: Carries complete sequence of Autonomous Systems (AS-PATH).

Loop Prevention Rule: If receiving BGP speaker detects its own AS in AS-PATH, discard route immediately!

Policy Routing: Routes chosen by commercial contracts & security policies rather than raw latency.

2. Hierarchical Routing Scaling

Routing Model	Table Entries	Update Frequency	Scalability Limit
Flat Routing	$O(N)$ (One per router)	Floods entire graph	Crashes at > 10,000 routers
Hierarchical (2-Level)	$O(\ln N)$ (Optimal Kamoun-Klein)	Localized to region	Scales to millions of Internet nodes